

GitHub Pages Research Project Guide

How to build, replicate, and organise research project websites
like **sidathc.github.io** — for free

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This guide covers:

- What technology powers your current site and why it works
- How to create an identical new project website from scratch
- How to turn **sidathc.github.io** into a central portfolio hub
- How to link all your projects together — fully free on GitHub Pages
- Tips for writing posts, adding images, and customising the look

Part 1: How Your Current Site Works

Your site at **sidathc.github.io** (currently the Contrails Australia project) is built with three free tools that work together seamlessly.

1.1 The Technology Stack

Tool	Role	Cost
GitHub Pages	Hosts your site for free, straight from a GitHub repository	Free
Jekyll	Static site generator — converts Markdown posts into HTML pages	Free
Minima theme	Clean, professional layout (nav, footer, post cards, RSS)	Free

1.2 How the pieces connect

When you push a Markdown file to the `_posts/` folder of your GitHub repository, GitHub automatically runs Jekyll to build HTML, then serves it at your domain. No server to rent, no build process to manage — just write and push.

1.3 Your repository structure

```
sidathc.github.io/
■■■ _config.yml      ← site settings (title, author, theme, plugins)
■■■ index.md         ← homepage (hero image + post list)
■■■ about.md         ← about page
■■■ _posts/          ← one Markdown file per blog post
■   ■■■ YYYY-MM-DD-slug.md
■■■ _layouts/        ← HTML templates (home.html, post.html)
■■■ _includes/       ← reusable snippets (navlinks, sharelinks, head)
■■■ css/
■   ■■■ override.css ← all custom styles (colours, cards, lightbox...)
■■■ js/              ← highlight.js for code blocks
■■■ assets/
    ■■■ favicon.svg
    ■■■ images/      ← all post images go here
```

1.4 How a post is structured

Each post in `_posts/` is a Markdown file with a small block of metadata at the top called *front matter*:

```
---
preview_image: my_chart.png      ← thumbnail shown on homepage card
excerpt: One-sentence teaser shown on the homepage card.
---

## Entry 7: Your Post Title

Write the post body in Markdown here.

![Chart caption](/assets/images/my_chart.png)
```

■ **Tip:** The file name must follow the pattern YYYY-MM-DD-entry-N-short-slug.md. The date controls ordering; the slug becomes the URL.

Part 2: Creating a New Research Project Website

Follow these steps to create a second (or third...) research site that looks and functions identically to the Contrails Australia site.

Step 1 — Create a new GitHub repository

- 1 Go to **github.com** and sign in as **sidathc**.
- 2 Click the **+** → **New repository**.
- 3 Name it after your project, e.g. **my-project-name**. Keep it public (required for free GitHub Pages).
- 4 Check **"Add a README file"**, then click **Create repository**.

Step 2 — Enable GitHub Pages

- 1 Inside the new repository, go to **Settings** → **Pages**.
- 2 Under **Source**, choose **"Deploy from a branch"**.
- 3 Select branch **main**, folder **/ (root)**, and click **Save**.
- 4 Your site will be live at **https://sidathc.github.io/my-project-name/** within a minute.

Free URL pattern: Every repository you own gets a free subdirectory URL:
github.com/sidathc/repo-name → sidathc.github.io/repo-name

The special repo **sidathc.github.io** lives at the root (no subdirectory).

Step 3 — Copy your site's template files

Clone both the existing site and the new repo, then copy the template:

```
# Clone your existing template site
git clone https://github.com/sidathc/sidathc.github.io template-site

# Clone the new project repo
git clone https://github.com/sidathc/my-project-name

# Copy the template skeleton into the new project
cp template-site/_config.yml      my-project-name/
cp template-site/index.md        my-project-name/
cp template-site/about.md        my-project-name/
cp -r template-site/_layouts     my-project-name/
cp -r template-site/_includes    my-project-name/
cp -r template-site/css          my-project-name/
cp -r template-site/js           my-project-name/
cp -r template-site/assets       my-project-name/
mkdir -p my-project-name/_posts
```

Step 4 — Customise _config.yml for the new project

Open my-project-name/_config.yml and change these lines:

```
title: "My New Project"
author: "Sidath Chandrasena"
description: "A short description of this research project"
email: "sidath.chandrasena[at]gmail.com"
github_username: "sidathc"
linkedin_username: "sidath-chandrasena"
```

```
# Important: tell Jekyll your site lives in a subdirectory
baseurl: "/my-project-name"      # ← add this line
url: "https://sidathc.github.io"
```

Note on baseurl: The root site (sidathc.github.io) does NOT need baseurl. Only add it for subdirectory project sites like /my-project-name.

Step 5 — Update index.md and about.md

Edit index.md — replace the hero image and welcome text:

```
---
layout: home
---

<div class="hero-banner" style="background-image: url('/my-project-name/assets/images/hero.jpg');"></div>
<p class="photo-credit">Photo by Author</p>

## Welcome!
Brief description of this research project's goals.
```

Edit about.md — update the project context, research questions, and your author bio. Keep the same References section format.

Step 6 — Add a hero image

- 1 Find a free image on **Unsplash** (unsplash.com) — download the full size.
- 2 Rename it to something descriptive (e.g. hero.jpg).
- 3 Place it in my-project-name/assets/images/.
- 4 Update the background-image URL in index.md as shown above.

Step 7 — Write your first post

Create a file in _posts/ following the naming convention:

```
# File: _posts/2026-04-05-entry-1-your-post-subtitle.md

---
preview_image: entry_1_chart.png
excerpt: One-sentence summary shown on the homepage card.
---

## Entry 1: Your Post Subtitle

Post body in Markdown...
```

Step 8 — Push to GitHub

```
cd my-project-name
git add .
git commit -m "Initial site setup"
git push origin main
```

GitHub Pages will build and deploy automatically within 1–2 minutes. Check the **Actions** tab in your repository to watch the build.

Part 3: Turning sidathc.github.io into a Portfolio Hub

Currently sidathc.github.io is dedicated to Contrails Australia. You can convert it — or keep it as-is and add a navigation hub — so visitors can find all your research projects in one place.

3.1 Option A (Recommended): Separate the hub from the projects

Keep each project at its own URL and make sidathc.github.io the hub:

- sidathc.github.io → Portfolio home listing all projects
- sidathc.github.io/contrails-australia/ → Contrails Australia site
- sidathc.github.io/new-project/ → New project site

Why this works: GitHub gives you unlimited free subdirectory sites from one account. Each project stays self-contained, has its own posts and styles, and can be worked on independently. The hub simply links to each one.

3.2 Migrating Contrails Australia to a subdirectory

To free up the root sidathc.github.io for the portfolio hub:

- 1 Create a new repo called **contrails-australia** on GitHub.
- 2 Copy all current files from sidathc.github.io into it.
- 3 Add baseurl: `"/contrails-australia"` to its `_config.yml`.
- 4 Enable GitHub Pages on the new repo. It will live at sidathc.github.io/contrails-australia/.
- 5 Now repurpose the sidathc.github.io repo as the hub (see section 3.3).

Note: If you already have a separate `contrails-australia` repo, you may simply need to add `baseurl` to its `_config.yml` and enable GitHub Pages on it.

3.3 Building the portfolio hub (sidathc.github.io)

Replace the current `index.md` in sidathc.github.io with a project gallery. Below is a complete example you can paste in:

```
---
layout: home
---

<link href="/css/override.css" rel="stylesheet">

<div class="hero-banner" style="background-image: url('/assets/images/hub-hero.jpg');"></div>

## Research Projects

I'm Sidath Chandrasena – software engineer and researcher exploring data-driven
questions in aviation, environment, and beyond.

<div class="project-grid">
```

```

<div class="project-card">
  <a href="https://sidathc.github.io/contrails-australia/">
    
  </a>
  <h3><a href="https://sidathc.github.io/contrails-australia/">Contrails Australia</a></h3>
  <p>Examining contrail formation in Australian skies using the Google Contrails API,
    OpenSky Network, and geospatial analysis.</p>
</div>

<div class="project-card">
  <a href="https://sidathc.github.io/new-project/">
    
  </a>
  <h3><a href="https://sidathc.github.io/new-project/">New Project Title</a></h3>
  <p>Short description of the second research project goes here.</p>
</div>

</div>

```

3.4 CSS for the project grid

Add these styles to `css/override.css` in the hub repo:

```

.project-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(260px, 1fr));
  gap: 1.6em;
  margin: 1.5em 0 2em;
}

.project-card {
  background: #ffffff;
  border: 1px solid #e4ecf5;
  border-radius: 8px;
  overflow: hidden;
  box-shadow: 0 2px 8px rgba(0,0,0,0.07);
  transition: box-shadow 0.2s;
}

.project-card:hover {
  box-shadow: 0 4px 16px rgba(0,0,0,0.13);
}

.project-card img {
  width: 100%;
  height: 160px;
  object-fit: cover;
  display: block;
  cursor: default;
}

.project-card h3 {
  margin: 0.8em 1em 0.3em;
  font-size: 1.05em;
}

```

```

}

.project-card p {
  margin: 0 1em 1em;
  font-size: 0.88em;
  color: #555;
  line-height: 1.5;
}

```

3.5 Option B: Keep sidathc.github.io as Contrails and add a nav link

If you prefer not to restructure, add a simple 'Projects' navigation page. Create `projects.md` in `sidathc.github.io`:

```

---
layout: page
title: All Projects
permalink: /projects/
---

### Contrails Australia
*You are here.* This site explores contrail formation in Australian skies.

---

### [New Project Title](https://sidathc.github.io/new-project/)
Short description. → [Visit site](https://sidathc.github.io/new-project/)

```

Jekyll's Minima theme will automatically add 'All Projects' to the navigation bar.

Part 4: Writing Posts — Reference Sheet

4.1 Markdown quick reference

Markdown	Result
<code>## Heading 2</code>	Large section heading
<code>### Heading 3</code>	Sub-section heading
<code>**bold text**</code>	Bold text
<code>*italic text*</code>	Italic text
<code>`inline code`</code>	Inline code
<code>```python ... ```</code>	Fenced code block with syntax highlight
<code>![Alt text](/assets/images/x.png)</code>	Inline image
<code>[Link text](https://...)</code>	Hyperlink
<code> Col1 Col2 </code>	Table (needs header + separator row)
<code><sup>[[1]](#ref1)</sup></code>	Footnote reference

4.2 Adding images to a post

- 1 Place the image file in `assets/images/` in your repo.
- 2 Reference it in Markdown: `![Caption text](/assets/images/filename.png)`
- 3 For project-specific repos (with baseurl) use:
`![Caption](/my-project-name/assets/images/filename.png)`
- 4 Set it as the homepage preview card thumbnail by adding `preview_image: filename.png` to the post's front matter.

Image sizing tip: The homepage card thumbnail is displayed at 175×120 px. For best quality, use images at least 350×240 px and keep file sizes under 500 KB.

4.3 Using the new-post helper script

Your repo already contains `new-post.sh` which auto-numbers posts:

```
cd sidathc.github.io    # (or your project repo)
./new-post.sh "Your Post Subtitle Here"
# Creates: _posts/2026-04-05-entry-7-your-post-subtitle-here.md
```

4.4 Front matter options

```
---
preview_image: filename.png    # thumbnail on homepage (optional)
excerpt: >                     # teaser text on homepage (optional)
  A one or two sentence
```

```
summary of this post.
author: Sidath Chandrasena    # overrides site-wide author (optional)
tags: [Data, Aviation]       # tags (optional)
---
```

4.5 References section (reuse this template)

```
---
### References

<div class="references">
  <p><a name="ref1">[1]</a> Author – <a href="https://example.com">Title</a></p>
  <p><a name="ref2">[2]</a> Author – <a href="https://example.com">Title</a></p>
</div>
```

Part 5: Customising the Look

All visual customisation lives in `css/override.css`. The Minima theme base styles are inherited automatically; you only override what you want to change.

5.1 Colour scheme

The key colours used across both the Contrails site and this guide:

Variable / Use	Hex value	Where used
Background	#f0f5fb	Page background, info boxes
Primary (navy)	#1a4a7a	Site title, headings, table headers
Link / accent (blue)	#2a6fad	Active nav, links
Border	#e4ecf5	Card borders, table lines
Text (muted)	#666 / #555	Excerpts, captions

To change the colour scheme for a new project, open `css/override.css` and do a find-and-replace on the hex values above.

5.2 Changing the site font

The site title and nav use **Space Grotesk** (loaded from Google Fonts). To switch fonts, edit the two lines in `_includes/head.html`:

```
<!-- In _includes/head.html -->
<link href="https://fonts.googleapis.com/css2?family=YOUR+FONT:wght@700&display=swap" rel="stylesheet"

/* In css/override.css */
.site-title, .site-title:visited, .page-link {
  font-family: 'Your Font', sans-serif;
}
```

5.3 Hero banner height and position

```
.hero-banner {
  height: 220px; /* increase for taller hero */
  background-position: center 70%; /* adjust vertical focus */
}
```

5.4 Post card thumbnail size

```
.post-preview-thumb { flex: 0 0 175px; }
.post-preview-thumb img { width: 175px; height: 120px; }
```

5.5 Adding Google Analytics (optional, free)

Add your tracking ID to `_config.yml`:

```
google_analytics: G-XXXXXXXXXX
```

Part 6: Complete Setup Checklist

New project site

- Create GitHub repository (public)
- Enable GitHub Pages in Settings → Pages
- Copy template files from sidathc.github.io
- Update `_config.yml` (title, description, baseurl)
- Replace `index.md` (hero image, welcome text)
- Update `about.md` (project context, author bio)
- Add hero image to `assets/images/`
- Write Entry 1 post in `_posts/`
- Push to GitHub and verify build in Actions tab
- Confirm site loads at sidathc.github.io/repo-name/

Portfolio hub (sidathc.github.io)

- Update `index.md` with project grid
- Add `.project-grid` CSS to `css/override.css`
- Add thumbnail image for each project to `assets/images/`
- Test all links point to the correct project URLs
- Optionally add an `/about/` page for the hub
- Push to GitHub and verify

For each new post

- Run `./new-post.sh 'Post Subtitle'` to create the file
- Add `preview_image` and `excerpt` to front matter
- Write post content in Markdown
- Place all images in `assets/images/`
- Push to GitHub — site updates within 60 seconds

Free resources

- **Unsplash** (unsplash.com) — free high-quality hero photos
- **Google Fonts** (fonts.google.com) — free web fonts
- **GitHub Pages docs** — docs.github.com/en/pages
- **Jekyll docs** — jekyllrb.com/docs

- **Minima theme** — github.com/jekyll/minima
- **Markdown guide** — markdownguide.org/basic-syntax

Summary of free URLs you can create:

`sidathc.github.io` → portfolio hub

`sidathc.github.io/contrails-australia/` → Contrails Australia

`sidathc.github.io/<any-repo-name>/` → any future project

Each is free, independently deployable, and shares the same look and feel.